

# MPEG Microbiome Decoding the Dialogue Challenge

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## Overview

### Track 1: Cytokine Prediction from Microbiome Data

The human body consists of highly dynamic microbial communities that participate in complex host-microbial interactions [1]. These organisms play important roles in maintaining normal physiological processes, one of which is modulating the immune system through cytokine production. Similarly, they have also been implicated in some host-related diseases such as inflammatory bowel disease (IBD), cardiovascular diseases and the type-2 diabetes mellitus. Recent advances in high-throughput sequencing and multi-omics profiling have generated vast, high-dimensional datasets that capture microbial community composition and function at unprecedented resolution. One of the most commonly used methods to analyze the microbiome is amplicon sequencing which involves the use of reads of specific taxonomic marker genes which comprises the use of evolutionarily conserved 16S rRNA gene [1].

Translating these complex interactions into valuable predictive insights about host immune responses remains a major challenge. Machine learning (ML) approaches offers a powerful solution into uncovering complex host-microbiome associations and identifying predictive microbial taxa or functional pathways [2].

### Aim

- To predict cytokine levels based on the microbiome composition.

## Methodology

- Divided read sequences in each sample into 8-sequence long and aggregated their occurrence
- Feature selection
- Modelling using an ensemble of three Penalised Linear Regression models

## Data Preparation

Dataset was obtained from 16S rRNA data stored in Fastq files

### Data Cleaning

- Removed invalid bases (bases with Ns, if present)
- Removed bases with low quality scores (below 20) and reads with sequence lengths less than 50
- Created 8-kmer sequences, with no successive skips.
- Aggregated them into counts
- Selecting samples with matching cytokine levels.

## Feature Engineering

### Normalisation

To prevent sample bias due to read length and depth, I normalised counts using the centred-log ratio method (CLR).

Centered Log-Ratio:

$$\text{CLR} = \log(1 + \text{counts}) - \text{mean}(\log(1 + \text{counts}))$$

### *Addition of covariates*

During data exploration, it was discovered that there was a difference between cytokine levels and some of the metadata. Age and BMI were converted into groups 5 and 4 groups, respectively. The metadata included in the model were fasting plasma glucose class (FPG) indicating the participant’s health status. Others include the BMI, Age group and body site. To capitalise on the effect of group interactions rather than their single effects, interaction terms of each group were added to the model. These include BMI x Age, BMI x FPG, Age x Body site, Age x FPG, and Body site x FPG

- Numerical variables were standardised while the categorical variables were one-hot/dummy encoded.

### *Feature selection*

Due to the huge size on the number of possible kmers to be generated from an 8kmer sequence ( $4^k$ ), feature selection was done to remove “irrelevant” features. Variance thresholding method was applied to remove CLR-normalised kmer counts below a cutoff. Features meeting this criteria were selected for modelling. This resulted in almost 80% trim of the total 65,536 unique 8-kmer sequences. Another level of feature selection (model-based feature selection) was done. An elastic net from Scikit-Learn (**MultiTaskElasticNet**) was used. While keeping the balance between l1 and l2 penalties, the l1-ratio of 0.5 was kept constant, while the alpha hyperparameter was tuned using a 5-fold cross-validation. The 1520 features left were further used for modelling.

### *Data splitting and cross-validation*

Data was randomly split into 80% train and 20% validation sets. To estimate model performance on unseen data, a 5-fold cross-validation approach was applied and results represented in RMSE (mean  $\pm$  sd), and a 95% confidence interval.

## Modelling

Three model architectures were developed and compared. They comprise a regularised linear regression models, and two regularised non-linear neural network models with one and two layers. The rectified linear unit (relu) was used as activation function while batchnormalisation and dropout layers were added to prevent overfitting. Additionally, to prevent overfitting, the L1 and L2 regulation methods were applied as well as the addition of dropout layers to model architectures.

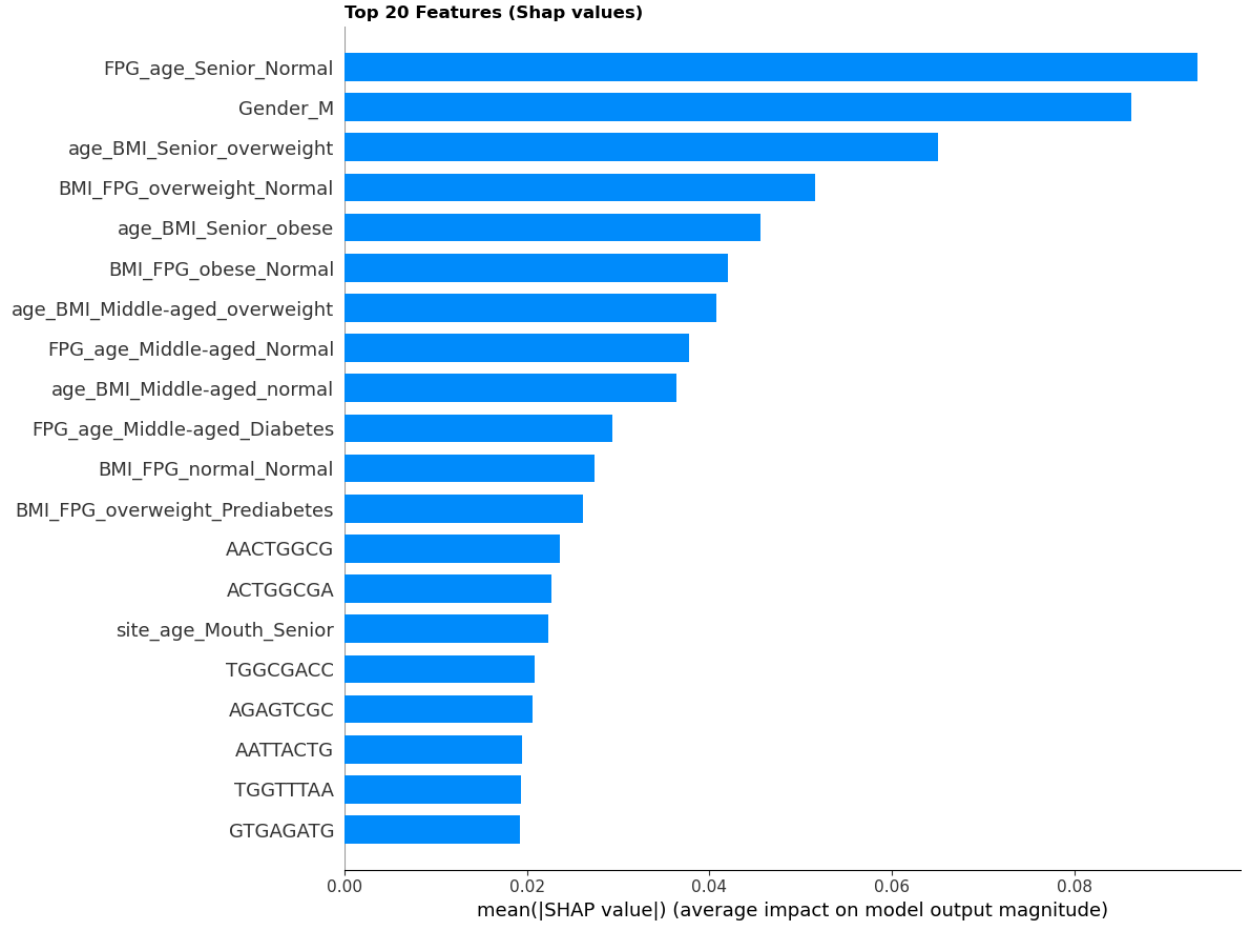
## Results

*Table 1: Model Performance on validation set*

|      | Linear  | Non-linear (One-layer) | Non-linear (Two-Layer) |
|------|---------|------------------------|------------------------|
| RMSE | 160.178 | 153.607                | 149.394                |

*Table 2: Model 5-fold cross-validation performance*

| Model                    | RMSE               | 95% CI           |
|--------------------------|--------------------|------------------|
| Linear Regression        | 155.824 $\pm$ 2.77 | 155.746, 163.144 |
| Non-Linear (Two layer)   | 145.51 $\pm$ 2.04  | 143.75, 149.     |
| Non-Linear (Three-layer) | 139.30 $\pm$ 5.41  | 133.78, 146.97   |



**Figure 1: Top 20 features (overall)**

**Table 3: Top 20 features (coefficients)**

|                                | Magnitude | Percentage dysregulated |
|--------------------------------|-----------|-------------------------|
| age_BMI_Senior_overweight      | 0.207     | 0.26                    |
| FPG_age_Senior_Normal          | 0.190     | 0.09                    |
| Gender_M                       | 0.179     | 0.38                    |
| age_BMI_Senior_normal          | 0.168     | 0.50                    |
| FPG_age_Senior_Prediabetes     | 0.163     | 0.26                    |
| age_BMI_Middle-aged_overweight | 0.153     | 0.71                    |
| FPG_age_Middle-aged_Diabetes   | 0.151     | 0.77                    |
| age_BMI_Middle-aged_normal     | 0.135     | 0.32                    |
| BMI_FPG_overweight_Prediabetes | 0.134     | 0.67                    |
| BMI_FPG_obese_Prediabetes      | 0.133     | 0.33                    |
| AACTGGCG                       | 0.133     | 0.15                    |
| ATTCTTCC                       | 0.132     | 0.88                    |
| TTGAGAAA                       | 0.131     | 0.91                    |
| ACTGGCGA                       | 0.130     | 0.17                    |
| site_age_Nasal_Adult           | 0.13      | 0.98                    |
| TCAGATAA                       | 0.127     | 0.83                    |
| BMI_FPG_normal_Prediabetes     | 0.127     | 0.45                    |
| age_BMI_Senior_obese           | 0.125     | 0.30                    |
| CTGGCGAC                       | 0.125     | 0.181                   |

|                       | Magnitude | Percentage dysregulated |
|-----------------------|-----------|-------------------------|
| BMI_FPG_normal_Normal | 0.124     | 0.45                    |

To visualise features with the most influence in predicting cytokine levels, shap values generated from the linear regression predictions were generated. To visualise their overall influence across all cytokines, absolute shap values from validation data were aggregated and visualised (Figure 1). From Figure 1, we see that interaction terms were the top features. To quantify the overall magnitude of cytokine regulation, the absolute values of these coefficients were taken and averaged across all cytokines. In addition, the proportion of cytokines showing positive versus negative coefficients were calculated, which we interpret as the percentage of cytokines that were upregulated or downregulated relative to the reference. This allowed us to summarize both the average strength of dysregulation and the breadth of cytokines affected and selected the top 20 based on their magnitudes (Table 3). Overweight senior citizens had less dysregulated cytokines than any other group. Their normal weight counterpart had equal dysregulated and upregulated cytokine levels. On the other hand, Senior citizens with normal fasting plasma glucose had higher cytokine levels compared to any other group.

On the other hand, gene sequences with **TGG**, **TGA** and **TCG** codons were amongst the top 8-kmer sequences. This may indicate that they significant role in microbial gene expressions. **TGA**, for example, normally functions as a stop codon, however, in some microbes, it is reassigned to encode selenocysteine which is important for enzymes involved in redox reactions and oxidative stress defence [3].

## References

1. Zhou, X., et al., (2024). Longitudinal profiling of the microbiome at four body sites reveals core stability and individualised dynamics during health and disease. *Cell Host & Microbe* 32, 506–526. <https://doi.org/10.1016/j.chom.2024.02.012>
2. Hernández Medina, R., Kutuzova, S., Nielsen, K. N., Johansen, J., Hansen, L. H., Nielsen, M., & Rasmussen, S. (2022). Machine learning and deep learning applications in microbiome research. *ISME Communications*, 2(1). <https://doi.org/10.1038/s43705-022-00182-9>
3. Jin, Y., Pu, D., Guo, H., Deng, Z., Chen, L., & Guo, F. (2022). T-G-A Deficiency Pattern in Protein-Coding Genes and Its Potential Reason. *Frontiers in Microbiology*, 13, 847325. <https://doi.org/10.3389/fmicb.2022.847325>